

DigiManuf

AI-Powered Digital Twin Ecosystem for Smart Manufacturing

—

Team: ML Mavericks

Challenges in Distributed Manufacturing

/02

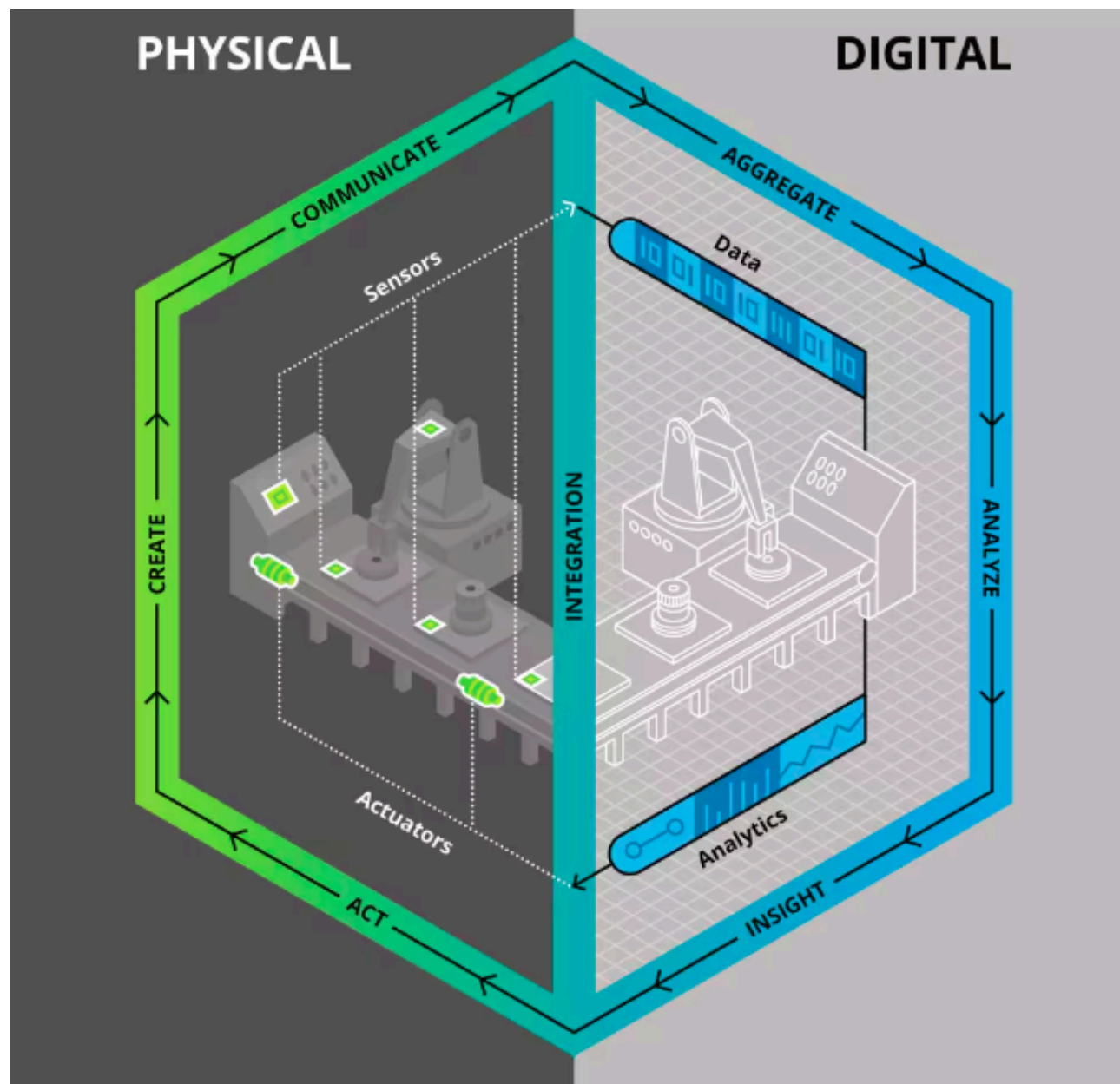


- Unexpected downtime and production losses
- Inefficient monitoring of geographically dispersed units
- Suboptimal workload distribution
- Lack of agility in responding to disruptions
- Ineffective quality control and safety management
- Limited data-driven decision-making capabilities
- Difficulty in predicting and mitigating equipment failures
- Inefficient energy usage and sustainability concerns

Introduction to DigiManuf

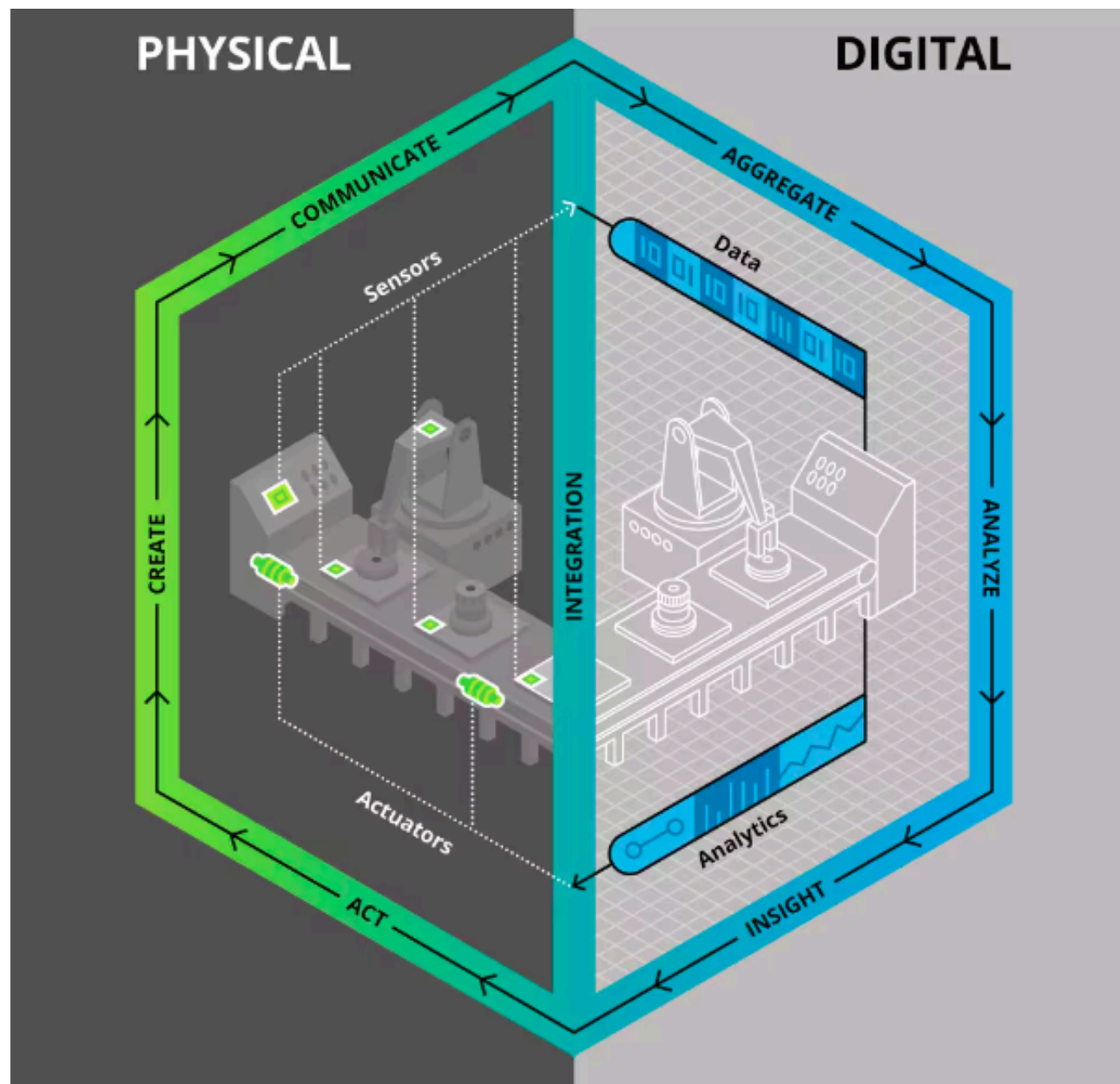
DigiManuf is a comprehensive AI-powered digital twin ecosystem that revolutionizes distributed manufacturing by:

- Creating virtual replicas of entire manufacturing operations
- Enabling real-time monitoring and control
- Leveraging AI for predictive maintenance and optimization
- Enhancing decision-making through data-driven insights
- Improving agility and responsiveness to market demands

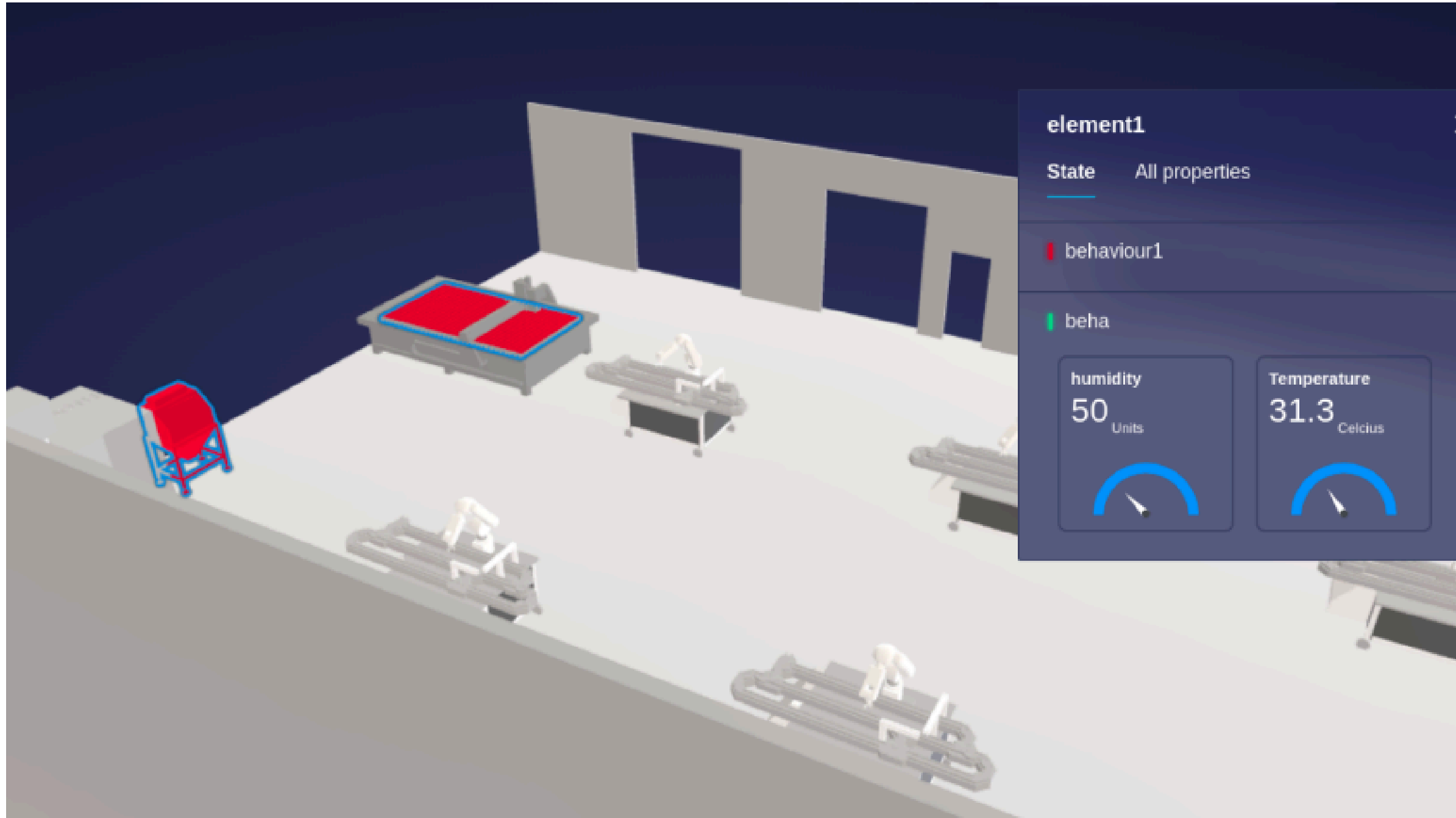


Powered by Cutting-Edge Technology

/04



- Digital Twin: Creating accurate virtual representations
- Artificial Intelligence: Driving predictive analytics and anomaly detection
- Internet of Things (IoT): Connecting and monitoring real-world assets
- Cloud Computing: Enabling scalable and accessible solutions
- Big Data Analytics: Processing vast amounts of manufacturing data
- Microservices: Enabling architectural approach where a large application is built as a set of small, independent services, each focusing on a specific business functionality.



Images of implemented Digital Twin

Key Components

/06

Advanced Digital Twin Technology

Real-time synchronization with physical assets

Interactive 3D visualizations

AI-Powered Anomaly Detection

Continuous monitoring of machine performance

Real-time alerts for potential issues

Predictive Maintenance

AI-driven analysis of historical and real-time data

Optimal maintenance scheduling

Interactive Control and Simulation

Adjust parameters through digital interface

Simulate changes before implementation

Intelligent Order Management

AI algorithms for optimal workload distribution

Dynamic reallocation based on machine availability

Adaptive Manufacturing

Real-time reallocation of tasks during disruptions

Ensure business continuity

AI-Powered Supply Chain Optimization

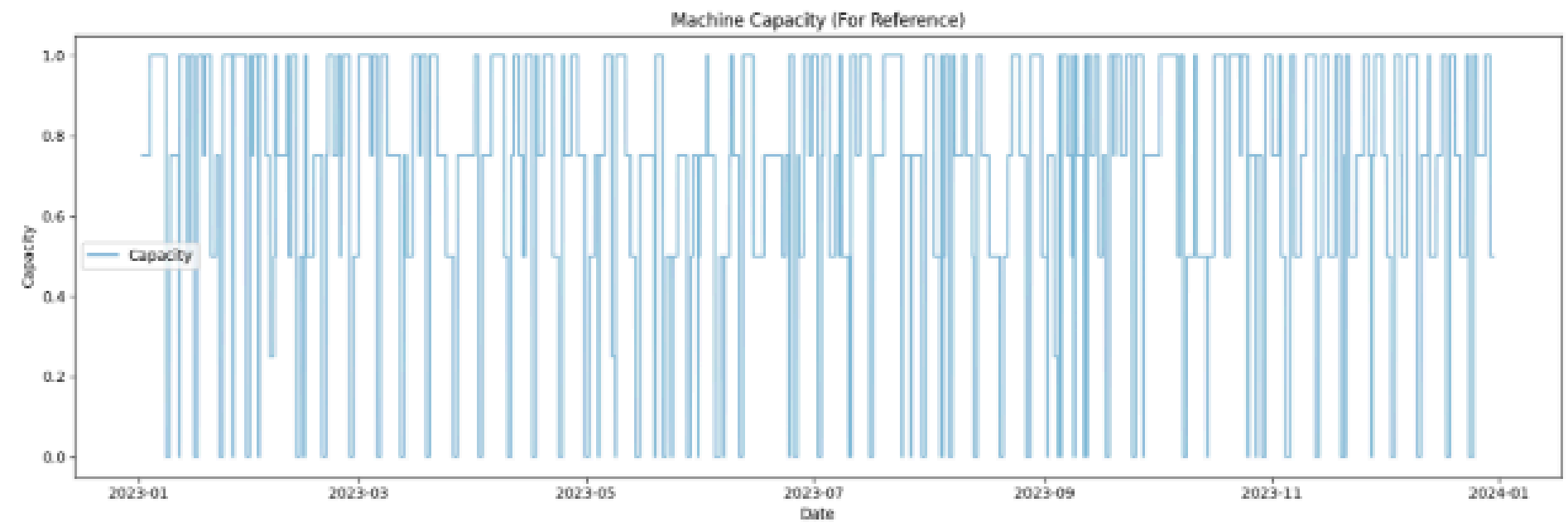
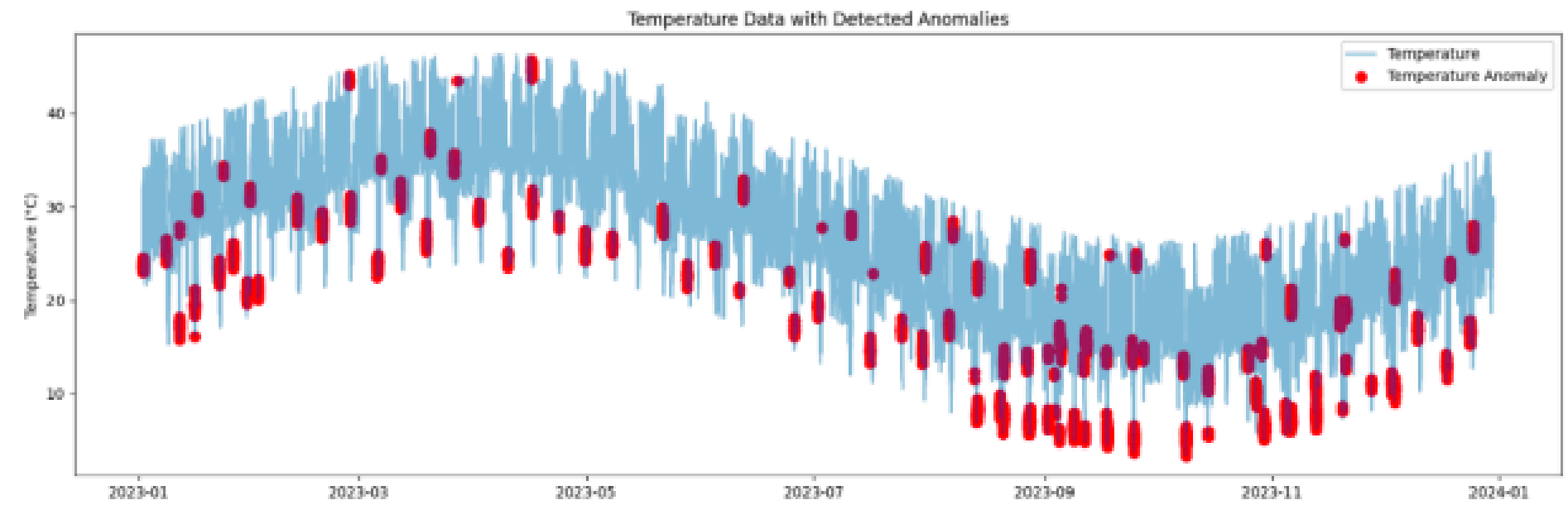
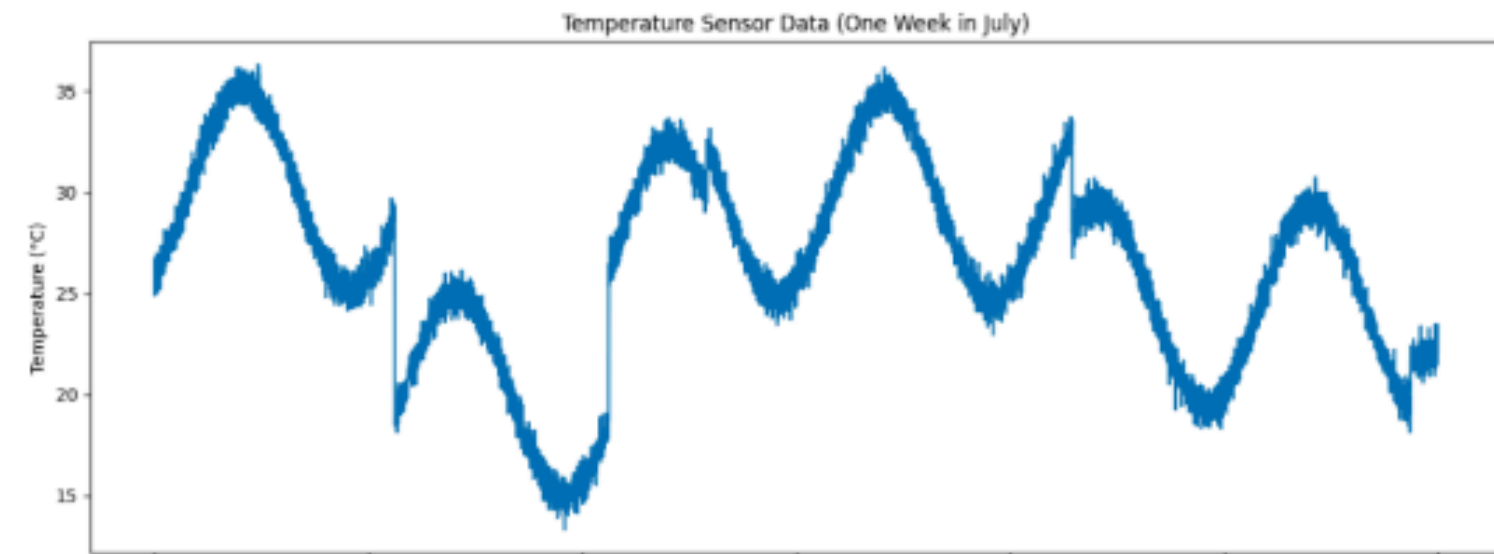
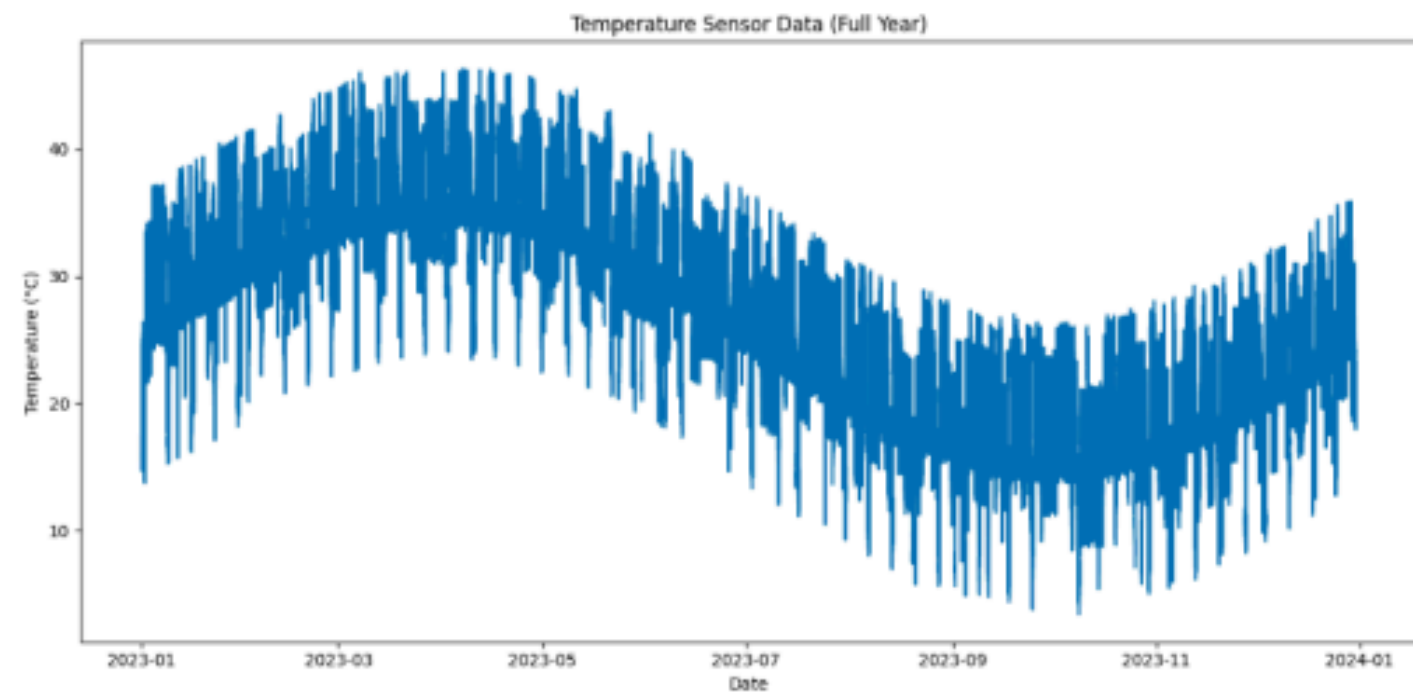
Comprehensive analysis of entire supply chain

Strategies to improve resilience

Computer Vision-Enhanced Safety and Quality Control

Real-time monitoring via camera feeds

Automated hazard and quality issue detection



Anomaly Detection & Value Prediction Graphs

How It Works?

- **Data Collection:** IoT sensors gather real-time data
- **Data Processing:** Cloud-based systems process incoming data
- **Digital Twin Update:** Virtual replica updated in real-time
- **AI Analysis:** Machine learning models analyze data and patterns
- **Insights Generation:** The system produces actionable insights
- **Control Implementation:** Changes applied to physical assets
- **Continuous Learning:** AI models update based on new data

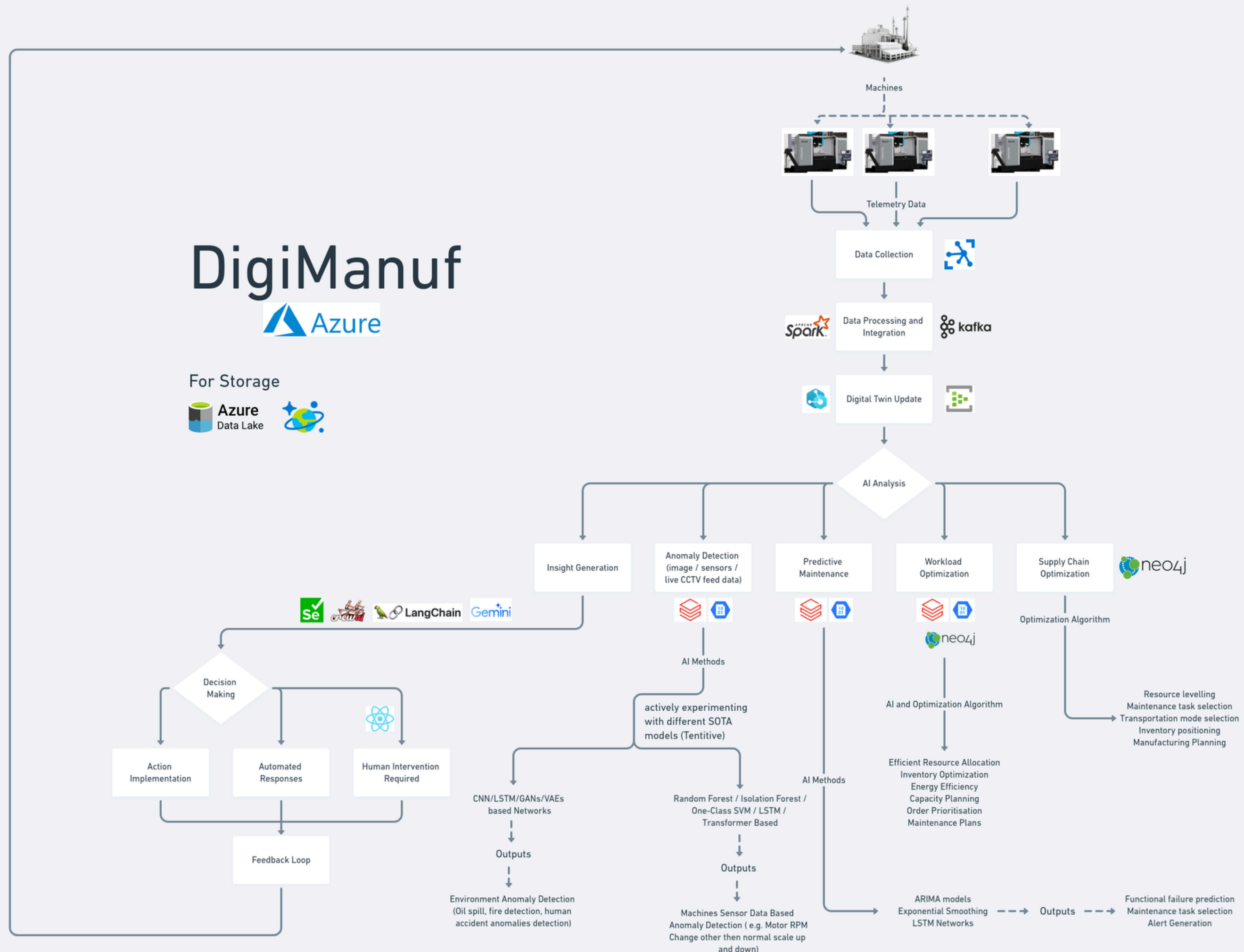
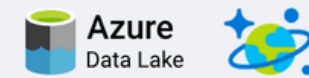
Challenges faced

- Limited Azure Credits
- Limited open-source code related to resource optimization
- Near to no manufacturing instrument sensor data available for the AI model training

DigiManuf



For Storage



Benefits and Impact

- Reduction in unplanned downtime
- Improvement in overall equipment effectiveness (OEE)
- Increase in productivity
- Lower maintenance costs
- Reduction in energy consumption
- Reduction in defect rates
- Faster time-to-market for new products
- Improvement in supply chain resilience

What Sets It Apart

- Comprehensive Integrations
- Advanced AI Implementation: Predictive and prescriptive analytics
- Real-time Adaptability: Dynamic resource allocation
- Scalability: Easily accommodate diverse manufacturing setups
- Sustainability Focus: Built-in metrics and optimization
- User-Centric Design: Intuitive interface for complex data

The Future with DigiManuf

/11



- Expansion to more industries (e.g., automotive, aerospace)
- Integration with emerging technologies (e.g., 5G, edge computing)
- Advanced simulation capabilities for product development
- Enhanced sustainability features and circular economy integration
- AI-driven autonomous manufacturing operations

Any Questions?